Spatio-Temporal Assessment of Urban Expansion Dynamics in Lahore District, Pakistan

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Abstract

Rapid urbanization is reshaping cities worldwide. Pakistan is urbanizing quickly, and Lahore, the country's second-largest city and a growing megacity with an estimated 14.8 million inhabitants in 2025 has seen strong population growth, loss of agricultural land, and recurring winter smog over the last decade. Satellite-based monitoring can help describe how and where the city is expanding. Using remote sensing techniques, a spatio-temporal assessment of urban expansion in Lahore District between 2000 and 2025 was done using ArcGIS Pro using both unsupervised and supervised image classification. Post-classification change detection was applied to calculate the area and percentage of land covers that converted to built-up. The study findings revealed that 59.11% (1022 km) of Lahore is today urban. Land use land cover (LULC) analysis of Lahore from 2000 to 2025 showed significant decreases in vegetation cover, water bodies, and barren land by 16.90%, 0.74%, and 25.57%, respectively, while built-up land increased by 43.23%. The study will support more evidence-based planning for compact, environmentally sustainable urban growth in line with Sustainable Development Goal 11 and ongoing planning initiatives in Lahore.

Keywords: change detection, urbanization, land use change, remote sensing, Lahore

1 Introduction

Rapid urbanization is a major worldwide phenomenon, drastically altering natural landscapes and creating huge barriers to sustainable development and climate change, especially in developing countries (Sahana et al., 2018; Sakieh et al., 2017). Pakistan exemplifies this trend, possessing one of the highest urbanization rates in South Asia (Karim et al., 2024). As Pakistan's second-largest and rapidly expanding metropolitan area in terms of population, the Lahore District has experienced substantial temporal and spatial changes due to this accelerated growth (Shrestha, 2021). Lahore has been projected to rise from the sixth fastest-growing city worldwide to the third fastest by 2030 (Karim et al., 2024). This highlights the importance of monitoring its urban dynamics.

The Land Use and Land Cover (LULC) of Lahore has drastically changed because of its sprawling growth, turning natural surfaces, permeable areas, and vegetation into impervious, built areas (Sattar et al., 2025). According to a study by Farhan et al. (2024), the expansion of urban areas in Lahore has resulted in a 2.33% loss of vegetation between 1990 and 2020. Also, it has been reported that built-up land has expanded along key road networks and radially outward from the Central Business District (CBD), particularly towards the south of the city (Sarwar & Mahmood, 2025). This accelerated urbanization has severe environmental

repercussions, notably the intensification of the Urban Heat Island (UHI) effect (Sattar et al., 2025). The reported rise in Land Surface Temperature (LST) is strongly correlated with increased built-up areas (Ahmad et al., 2023).

Monitoring changes in Land Use/Land Cover (LULC) is essential for successful urban planning and reaching sustainable development goals (Reynolds et al., 2017). This study specifically utilizes Remote Sensing (RS) and Geographic Information Systems (GIS) technologies to do a thorough spatiotemporal analysis of urban expansion over 25 years in Lahore District with spatial-temporal satellite imagery from the Landsat program. For long-term land cover change detection, the Landsat sensors provide the best possible balance between spatial (30 m) and temporal (16-day revisit time) resolution (Farhan et al., 2024). The study will also investigate how many other lands cover types, like vegetation, were converted to urban areas between 2000 and 2025. The results will support more evidence-based planning for compact, environmentally sustainable urban growth in line with Sustainable Development Goal 11 and ongoing planning initiatives in Lahore. The specific objectives of this study are to:

- Produce LULC maps of Lahore District for the years 2000 and 2025.
- Evaluate the thematic accuracy of the generated LULC maps.
- Compare the performance of unsupervised and supervised classification approaches.
- Quantify and analyze spatial patterns of urban expansion over the 25-year period.

2 Study Area

Lahore District, geographically located between 31°13′N to 31°43′N and 74°00′E to 74°39.5′E, is the provincial capital of Punjab province and Pakistan's second-largest metropolitan area. The city spans 1,772 square kilometres as it follows the left bank of the River Ravi through nine administrative towns and a cantonment zone which produces different urban areas (Sarwar et al.,2025). The first recorded population of Lahore in 1950 reached 835,769 residents but the city will experience 14.8 million projected residents by 2025 at a yearly population growth rate of 2.91% (World Population Review, 2025). The fast population growth has led to extensive urban development which expands into surrounding areas known as peri-urban zones and along transportation routes. These dynamics make Lahore a good case study for spatio-temporal urban expansion assessment for providing valuable insight into spatial growth patterns and land use change. Figure 1 shows the district's boundary, major road network, and its context within Punjab and Pakistan.

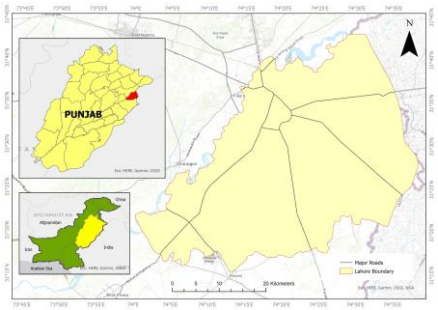

Figure 1. Map of Study Area: Lahore District, Pakistan

3 Data and Methods

3.1 Definition of Map Technical Specifications

The final outputs of this study consist of thematic land use/land cover (LULC) maps of Lahore District for the years 2000 and 2025, produced to support the analysis of long-term urban expansion. The maps were generated at the spatial resolution of the input satellite imagery and are intended for analysis at the district level. All maps were referenced to the WGS 84 coordinate system, projected to UTM Zone 43N, to ensure spatial consistency between temporal datasets. Since different geographic areas in Lahore district might contain similar land cover types, geographic stratification cannot improve accuracy and wasn't done. The classification scheme includes four land-cover classes: water, built-up areas, vegetation, and barren land / sparse vegetation. These were selected to represent the dominant land-use categories relevant to urban growth analysis. Thematic accuracy assessment was done to ensure the reliability of the produced maps for spatial interpretation and usage.

3.2 Data acquisition

3.2.1 Selection, acquisition and Pre-processing of satellite images

This study used two cloud-free spatial-spectral Landsat satellite images for the year 2000 and 2025 respectively. Landsat 5TM and Landsat 8OLI/TIRS (Path 149, Row 038), both of 30m resolution and obtained from the USGS earth explorer (http://earthexplorer.usgs.gov/) were used for the classification of land use/cover. Landsat Imagery was selected due to its suitability for district-scale urban studies and its historic archive of consistent temporal availability. The study was done using images in the same period (April) due to optimal weather conditions for data acquisition and the specific stage of vegetation cover, addressing a complex land-change problem. Preprocessing is an important part of satellite image interpretation; however, it was not done as the images were obtained as standards products, i.e., radiometrically and geometrically corrected.

3.2.2 Construction of the Ancillary Database

To support image interpretation, training sample selection, and thematic accuracy assessment, an ancillary database was compiled. It contained the administrative boundary of Lahore District, which was used for image sub-setting and spatial consistency across temporal datasets. High-resolution Google Earth imagery was employed as reference data for the identification and validation of land-cover classes. In addition, findings from previous land use/land cover studies conducted in the region were consulted to support class definition and interpretation of urban expansion patterns.

3.3 Data Processing and Preparation

3.3.1 Exploratory analysis of the Images

Using the RGB system, visual inspection of multiple band combinations enabled the identification of major land-cover types, including built-up areas, vegetation, bare land, and water bodies. To highlight built-up areas, the SWIR, NIR and Red (RGB) was used while the NIR, Red and Green (RGB) was used to highlight vegetation. This highlighted potential spectral confusion between built-up and bare land areas, particularly in peri-urban zones.

3.3.2 Definition of Information extraction strategy

This study used a combined classification approach by integrating unsupervised and supervised classification methods to exploit their complementary strengths. Unsupervised classification was first applied to explore spectral variability and support the identification of dominant land-cover patterns. Based on the result, supervised classification was performed to generate the final thematic maps using representative random training samples.

3.3.3 Band transformations

As an added step to improve visual interpretation and training sample selection, the Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI) and Normalized Difference Water Index (MNDWI) were used to improve the discrimination of vegetation, urban area and water bodies respectively.

3.4 Feature Selection and Classification

The study employed a pixel-based classification approach which classified satellite imaging pixels based on spectral reflectance properties that are similar. Thus, no image segmentation was performed. The feature set consisted of the original multispectral bands, which provide sufficient discriminatory power for the identification of major land-cover classes at the district scale. The classification process was applied independently to the 2000 and 2025 images to enable consistent spatial-temporal comparison.

For both Landsat images, an unsupervised classification approach was first employed using the ISO Cluster algorithm in ArcGIS Pro. A total of 15 natural occurring clusters were identified, inspected and merged into the four classes defined for the study. Subsequently, supervised classification was conducted using representative training samples for each land-cover class. A minimum of 50 training samples (freehand polygons) were created for each class. The supervised classification was done using Random Forest, Support Vector Machine and Maximum Likelihood algorithms. However, by visual inspection using the ancillary data, the Maximum Likelihood result was accepted to be evaluated.

3.5 Thematic accuracy assessment

The thematic accuracy of the classification was evaluated using an error matrix approach. This was done for both the unsupervised and supervised classification results. For each of the 2000 and 2025 images, 200 reference points were created using random sampling tools in ArcGIS Pro. These reference points were assigned to the correct classes using Google Earth Pro and support from the available ancillary data. The error matrix shows the accuracy of the classification. The columns represent the classes from the reference value, while rows correspond to the classes derived from the classified image. The diagonal cells describe pixels that were correctly detected, while the off-diagonal cells indicate a mismatch between reference and classed data. Accuracy metrics, including overall accuracy and class-specific user's and producer's accuracy, were computed to quantify classification performance.

3.6 Change Detection

To find areas of change, the post-classification comparison technique was employed. It involves classifying images and comparing the relevant classes. The post-classification comparison approach obtained the greatest classification accuracy in a comparative analysis of diverse techniques as seen in the work of Sun et al. (2009). The change in classes, specifically vegetation and barren, into built-up/urban was calculated using change detection wizard in

ArcGIS Pro. The post-classification comparison was conducted by converting the classified raster images into vector layers. The changes in LULC were calculated for each land cover type and represented in separate images in this study for better understanding.

4 Result

4.1 Land Cover Pattern of 2000: Unsupervised and Supervised

Figure 3 shows the land cover pattern of 2000 as obtained from the unsupervised (left) and supervised (right) classification. The supervised classification for 2000 exhibits a spatially coherent distribution of land-cover classes, with built-up areas clearly concentrated in the central urban core and along major linear features, vegetation dominating the peripheral and eastern zones, and barren or sparsely vegetated land occurring mainly in transitional areas. Water bodies are distinctly mapped along river and canal networks. In contrast, the unsupervised classification shows a more fragmented pattern, where built-up areas are mixed with barren land, particularly in peri-urban zones, and vegetation and barren classes display higher spectral mixing.

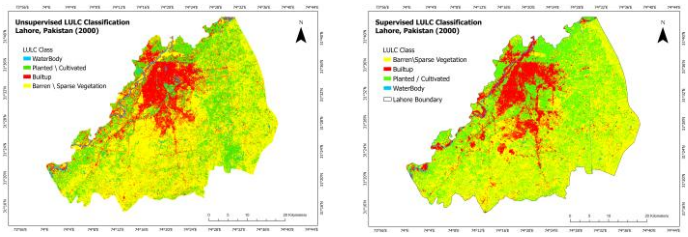

Figure 3. Land cover pattern for Lahore District for 2000: Unsupervised vs Supervised

4.2 Land Cover Pattern of 2025: Unsupervised and Supervised

For the unsupervised and supervised classification done (Figure 4), four classes of land use/land cover for 2025 were identified: water, vegetation, built-up and barren land. There appeared to be a dominant appearance of the built-up areas around the central and northern part of Lahore, which also spirals a bit towards the south. The vegetation class in the supervised are notably constrained compared to the vegetation class in the unsupervised map,

appearing as smaller discrete patches. In both classification techniques, water is identified in the north and northwest of Lahore.

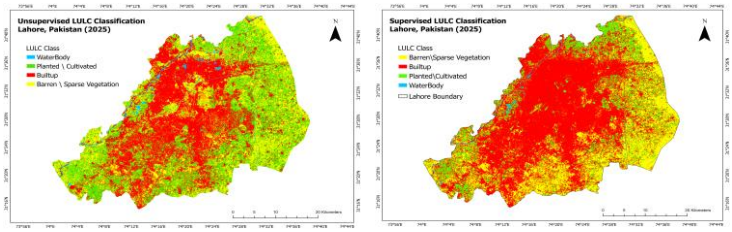

Figure 4. Land cover pattern for Lahore District for 2025: Unsupervised vs Supervised

4.3 Map Accuracy

4.3.1 Unsupervised maps

Table 1 shows the error matrix for the 2000 unsupervised classification. The classifier performed averagely with an overall accuracy of 56.5%. The user's and producer's accuracies of Barren/Sparse vegetation, built-up areas and Vegetation are 65.32% and 69.83%; 69.57% and 55.17%; and 32.00% and 30.77% respectively. Water was poorly classified with none of the reference water samples correctly identified. Most water pixels were misclassified as Vegetation and Barren, indicating strong spectral confusion, likely due to similar reflectance characteristics or mixed pixels. Table 2 shows the error matrix for the 2025 unsupervised classification with an overall accuracy of 54.0%, indicating a moderate but slightly lower performance compared to earlier classification results. Built-up areas, Barren areas, vegetation and water have a user's and producer's accuracies of 93.48% and 52.44%; 37.21% and 60.38%; 50.79% and 50.79%; and 20.00% and 50.00% respectively.

4.3.2 Supervised maps

Table 3 shows the error matrix for the 2000 supervised classification indicating a high overall accuracy of 71.5%, demonstrating improved classification performance compared to the unsupervised approach. Built-up areas, Vegetation, and Barren land were classified with a user's accuracy and a producer's accuracy of 86.67% and 83.87%; 73.33% and 63.95%; and 65.26% and 74.70% respectively. For the built-up areas, this indicates a strong agreement between classified and reference data while some confusion between vegetation and barren was observed. The error matrix for the 2025 supervised classification in Table 4 shows a high overall accuracy of 79.5%. The Built-up areas have a user's accuracy of 75.21% and a producer's accuracy of 100%, indicating that all reference urban pixels were correctly classified, despite some overclassification. Barren areas also exhibited strong reliability, with a user's accuracy of 87.76% and a producer's accuracy of 67.19%. Vegetation demonstrated good classification performance, achieving a user's accuracy of 82.76% and a producer's accuracy of 58.53%, though some confusion with barren areas and built-up areas were present. The Water class showed perfect user accuracy (100%) but low producer accuracy (25%), reflecting limited reference samples and underrepresentation. Overall, the results confirm that supervised classification significantly enhances classification accuracy and class separability for the study area in 2000 and 2025.

Table 1 Error Matrix of Unsupervised Classification – 2000.

	Water	Vegetation	Built-up	Barren	Total (User)	User Accuracy (%)
Water	0	1	0	2	3	0.00
Vegetation	1	16	0	35	52	32.00
Built-up	1	6	16	6	29	69.57
Barren	1	27	7	81	116	65.32
Total (Producer)	3	50	23	124	200	
Producer Accuracy (%)	0	30.77	55.17	69.83		
Overall Accuracy		0.565	0.565 * 100 = **56.5**			

Table 2 Error Matrix of Unsupervised Classification – 2025.

	Water	Vegetation	Built-up	Barren	Total (User)	User Accuracy (%)
Water	1	1	1	2	5	20.00
Vegetation	1	32	12	18	63	50.79
Built-up	0	2	43	1	46	93.48
Barren	0	28	26	32	86	37.21
Total (Producer)	2	63	81	53	200	
Producer Accuracy (%)	50	50.79	52.44	60.38		
Overall Accuracy		0.540	0.540 * 100 = **54.0**			

Table 3. Error Matrix of Supervised Classification – 2000.

	Vegetation	Built-up	Barren	Total (User)	User Accuracy (%)
Vegetation	55	2	18	75	73.33
Built-up	1	26	3	30	86.67
Barren	30	3	62	95	65.26
Total (Producer)	86	31	83	200	
Producer Accuracy (%)	63.95	83.87	74.70		
Overall Accuracy	0.715	0.715 * 100 = **71.5**			

Table 4 Error Matrix of Supervised Classification – 2025.

	Water	Vegetation	Built-up	Barren	Total (User)	User Accuracy (%)
Water	1	0	0	0	1	100
Vegetation	1	24	0	4	29	82.76
Built-up	2	11	91	17	121	75.21
Barren	0	6	0	43	49	87.76
Total (Producer)	4	41	91	64	200	
Producer Accuracy (%)	25	58.53	100	67.19		
Overall Accuracy		0.795	0.795 * 100 = **79.5**			

4.4 Spatiotemporal Variation in Land Cover

4.4.1 Changes in Other Land Cover to Urban

Change in vegetational land has been shown in Figure 5 (left). The green color area describes the unchanged vegetation area, which is about 258 km². The vegetation land to Built-up has been described by the red color, which area is about 292 km². Change in the Barren Land has been shown in Figure 5 (right). The Yellow color area describes the unchanged Barren Land, which is about 438 km². The Barren land to Built-up has been described by the red color, which is about 442 km².

4.4.2 Urban Expansion

The built-up area of Lahore district was 275 Km² in 2000. It increased to 1022 Km² in 2025, thus recording a growth of 43.23% (Table 5). While there is a loss of agricultural and barren land, 16.90% and 25.57% respectively (Figure 6). This rapid population growth is one of the major factors for urban expansion in the city of Lahore, and another reason for radical changes in the land use of Lahore. The major expansion took place along Ferozpur Road, Riawind Road, and Canal Road. From Figure 7, the light red represents urban areas in 2000 while the dark red represents urban areas in 2025. The expansion and physical growth of the city is observed in the southeast, south, and southwest directions. The key factor behind the urban expansion in these directions is the availability of open spaces & large parcels of land, connected roads, a pollution-free environment and accessibility, and possibilities for the residents who reside far away from the center of the city. Private and public housing complexes are being established in Lahore at present time in a disorganized and quick manner. The ribbon development along G.T. Road and Ferozepur Road resulted from this growing tendency. Lahore city had expanded into Aziz Bhatti Town, Data Ganj Baksh Town, Wagha Town, and Nishtar Town.

Table 5. Land cover statistics of Lahore 2000 and 2025

LandCover Indicator	2000		2025		Percentage Change
	Area (Sq.km)	%age	Area (Sq.km)	%age	
Barren/Sparse Vegetation	880	50.89	438	25.32	-25.57
Builtup	275	15.88	1022	59.11	43.23
Vegetation	550	31.84	258	14.94	-16.90
Water	24	1.37	11	0.63	-0.74

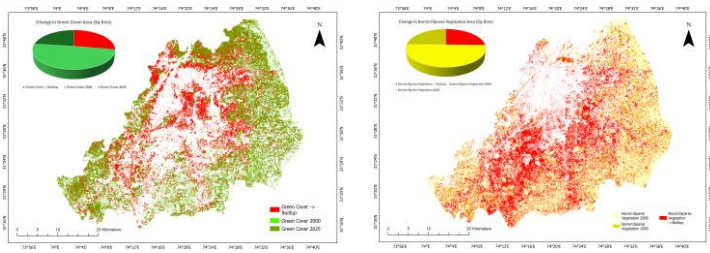

Figure 5. Left: Changes in Agriculture. Right: Changes in Barren/Sparse Vegetation

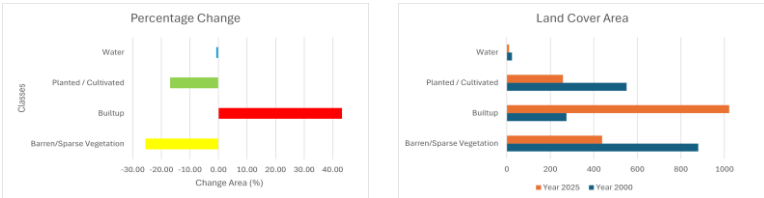

Figure 6. Left: Percentage changes in Land Cover. Right: Area of Land Cover

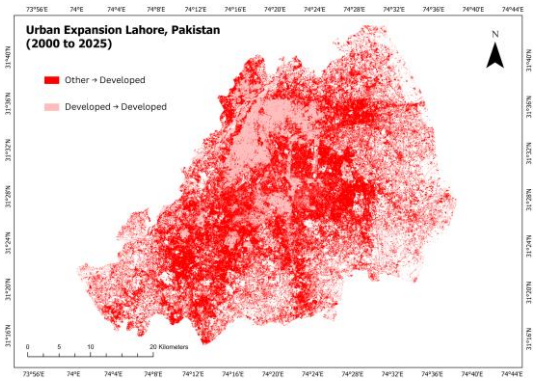

Figure 7. Urban Expansion of Lahore District from 2000 to 2025

5 Discussion

This study documented the progress of urban expansion in Lahore, Pakistan, from 2000 to 2025. Land use land cover (LULC) analysis of Lahore from 2000 to 2025 showed significant decreases in vegetation cover, water bodies, and barren land by 16.90%, 0.74%, and 25.57%, respectively, while built-up land increased by 43.23%. According to Karim et al. (2025), housing projects and infrastructure measures like metro bus and orange train are responsible for the rapid growth of Lahore's urban areas, and this growth is projected to increase from 53.6% urbanization in 2023 to 75.8% by 2048. Lahore's rapid urban expansion, spearheaded by the Lahore Development Authority and private developers, has come at a steep environmental cost. Research indicates that urban land cover rose from 25% in 2000 to 35% by 2015, while vegetation plummeted from 59% to just 34% (Shirazi & Kazmi, 2016; Farhat et al., 2018; Tariq et al., 2023). This shift, confirmed by NDVI and NDBI analyses, has intensified the Urban Heat Island (UHI) effect and placed unsustainable pressure on the city's infrastructure (Imran & Mehmood, 2020; Nasar-U-minallah et al., 2021). To mitigate these risks, Lahore must prioritize smart city strategies and sustainable planning to preserve dwindling agricultural land and enhance overall livability (Nadeem et al., 2021).

6 Conclusion

Lahore, a UNESCO city in northeast Pakistan, is experiencing growing urbanization, which poses ecological concerns and undermines the city's dedication to cultural preservation. High-resolution Landsat imagery was employed to study Lahore's urban growth from 2000 to 2025. The study findings revealed that 59.11% (1022 km) of Lahore is today urban. This increase is having a severe influence on ecosystems, resulting in a 0.74% decline in water bodies, a 16.90% loss of Vegetation, and a 25.57% decrease in barren areas. The study underlines the critical need for efficient urban planning to achieve SDG 11, asking planners to encourage smart city projects and protect green and blue areas. It can also be used as a model for other urbanized cities facing similar difficulties.

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